SAKARYA UNIVERSITY · COP31 GREEN ZONE

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yeşilkod

Measures software energy, fixes the waste, proves the gain.

How much electricity does this line of code use?

```
- return [no for no in GELENLER if no in kayitlilar]
+ kayitlilar_kume = set(kayitlilar)
+ return [no for no in GELENLER if no in kayitlilar_kume]
```

465× less energy, identical output.

At one million requests a day, this single line emitted about 41 kg CO_2e a year. The engine rewrote it by itself.

WHY IT MATTERS

Code has an invisible carbon cost. Yeşil Kod makes it visible.

Data-centre electricity use is expected to grow from **415 TWh** in 2024 to **~945 TWh** in 2030 (IEA). Part of that energy feeds waste nobody notices: loops that repeat the same work thousands of times, needless database queries, results computed over and over.

1 Find

Walks the syntax tree and flags thirty energy-wasting patterns, line by line.

2 Fix

Rewrites six of them itself: batched query, set lookup, loop hoisting, caching, min/max, copy removal.

3 Verify

Runs both versions and rejects the fix unless the outputs are identical.

4 Measure

Reads CPU energy counters and converts the gain to carbon on the Türkiye grid.

Green Coding and Digital Carbon Footprint Optimisation Tool
"Yeşil Kod" is Turkish for "Green Code".

yeşilkod

RESULTS

Four real-world wastes. The engine fixed all four.

Route count uncached double recursion → lru_cache	**~4,000×**
Attendance check list lookup inside a loop → set()	**465×**
Price list loop-invariant work → moved out of the loop	**51×**
Order report N+1 queries → one batched IN query	**4,001 → 2 queries**

Reduction in CPU energy per request. Estimated measurement mode, 26 October 2026. None of the optimised code was written by hand.

Carbon accounting

$$SCI = (E \times I + M) / R$$

- **E** measured CPU energy
- **I** Türkiye grid, 341 gCO_2e/kWh
- **M** embodied emissions (next phase)
- **R** per request

Honest limits

- Only CPU energy is measured; memory, network and cooling are excluded. The gain is a lower bound.
- Caches are cleared before every request.
- The engine currently supports Python.

SCAN ME

Try your own code on your phone.

The analysis runs on your phone; your code never leaves it. The site also has the film with English subtitles.

→ yes3760.github.io/yesil-kod
→ Source: github.com/yes3760/yesil-kod

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Sources: IEA, Energy and AI (2025) · Low-Carbon Power, Türkiye (Aug 2025–Jul 2026) · Green Software Foundation, SCI